Topic 13: Evaluating and Testing Prompts

13.1 Why Evaluation Matters

- A prompt that works once might fail 20% of the time on real-world inputs -- you won't know without testing
- Prompt evaluation is analogous to unit testing in software engineering
- Without evaluation: you ship a prompt and discover failures in production
- With evaluation: you catch regressions early, compare variants, and build confidence in changes
- Evaluation is especially critical for production systems where the model output triggers real actions

13.2 Building a Test Set

- Collect 20-100 representative real inputs that the prompt will process in production
- Include edge cases: empty inputs, very long inputs, inputs with unusual characters, adversarial inputs
- For each input, define the expected output or acceptance criteria
- Label a portion of the test set with ground-truth answers for automated scoring
- Refresh the test set periodically as real-world inputs evolve

13.3 Evaluation Metrics

- Exact match: output exactly equals expected -- use for classification, short answers
- F1 / precision / recall: use for extraction tasks (named entities, key facts)
- BLEU / ROUGE: n-gram overlap scores -- use for summarisation and translation
- LLM-as-judge: use a separate model call to score output quality -- scales well but adds cost
- Human evaluation: gold standard but slow and expensive -- use for ambiguous tasks
- Latency and cost: track tokens used and response time -- critical for production

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// LLM-as-judge example prompt:
You are an impartial evaluator.
Rate the following answer on a scale of 1-5 for:
  - Accuracy (does it answer the question correctly?)
  - Completeness (does it cover all key points?)
  - Clarity (is it easy to understand?)

Question: [original question]
Expected answer: [reference answer]
Model answer: [answer to evaluate]

Return a JSON: {accuracy: N, completeness: N, clarity: N, reasoning: '...'}
```

13.4 A/B Testing Prompts

- A/B test: run two prompt variants on the same inputs and compare quality scores
- Change one variable at a time: instruction wording, number of examples, temperature
- Minimum sample size: at least 50-100 inputs per variant for statistical significance
- Track: average quality score, failure rate (output doesn't meet criteria), latency, cost
- Adopt the winning variant, document what changed and why it improved performance